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ENGO 664: DATA ANALYSIS IN ENGINEERING

Statistical Testing and Variance Component Estimation

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ENGO 664 — Lab 4

Statistical Testing and Variance Component Estimation

Problem Statement

This laboratory extends the least-squares adjustment work from Lab 1 by focusing on statistical testing and variance component estimation. Two main data sets are analyzed:

1. **Double-differenced GPS range noise (Lab4Q1data.txt)** from a zero baseline, sampled every second. These measurements are theoretically uncorrelated noise and are used to investigate whether the noise follows a normal distribution.
2. **The angle–distance network from Lab 1** (three control points A, B, C and two unknown points P1, P2) with 12 angles and 6 distances and an a-priori stochastic model. This network is used to:
   - Detect and identify possible blunders using data snooping.
   - Analyze the correlation structure among identification statistics.
   - Remove identified blunders and estimate measurement variances.
   - Study the effect of introducing artificial errors of size 5σ and 3σ to selected distance measurements.

The specific questions are:

1. Design and perform a normality test on the GPS noise data with 95 % confidence.
2. For the Lab 1 network:
   - Q2.1: Perform a global variance-factor test and local data snooping (w-tests) at 95 % confidence.
   - Q2.2: Compute the correlation coefficients of the identification statistics and analyze their impact on blunder detectability.
   - Q3: If more than two blunders are detected, analyze possible causes.
   - Q4: Remove the two identified blunders, re-estimate measurement variances, and compare with manufacturer specifications.
   - Q5–Q7: Introduce artificial errors of 5σ and 3σ to selected distances (P2–C, and P1–A & P2–C), repeat the tests, and interpret the results.

Tools and Programming Language

All computations were implemented in Python 3.11 in a Jupyter Notebook environment, consistent with previous labs.

The following libraries were used:

- NumPy for array and matrix operations.
- SciPy (scipy.stats) for chi-square and normal quantiles and cumulative distribution functions.
- Matplotlib for plotting histograms, Q–Q plots, and heatmaps of correlation matrices.

The Lab 1 non-linear adjustment algorithm (Gauss–Newton with numerical Jacobian and proper angle wrapping) was reused through a helper function lab1_get_matrices(), which returns:

- Design matrix A, observation vector l, weight matrix P,
- Adjusted parameters $\hat{\boldsymbol{x}} = [\boldsymbol{xP1}, \boldsymbol{yP1}, \boldsymbol{xP2}, \boldsymbol{yP2}]^{T}$,
- Residuals $\hat{v}$, residual cofactor matrix $\boldsymbol{Q\_vv}$, and

- A-posteriori variance factor $\hat{\boldsymbol{\sigma}}_0^2$.

Data and Stochastic Model Used

Q1 — GPS DD Noise (Lab4Q1data.txt)

The file Lab4Q1data.txt contains double-differenced range measurements from a zero baseline GPS observation. Each record includes time tags and a double-differenced range value. Only the last column (range noise) is used for the statistical test.

Let (x_i) denote the 300-sample subset of DD noise values used in this lab. Empirical statistics were:

- Number of samples: $N = 300$
- Sample mean: $\bar{x} \approx 0.449m$
- Sample standard deviation: $s \approx 4.877m$

Theoretical assumption:

$xi{\sim}N(\mu,\sigma2)$ independent and identically distributed (i.i.d.).

Q2–Q7 — Lab 1 Survey Network

The network is identical to Lab 1: three control points A, B, C with known coordinates and two unknown points P1, P2. Twelve angles and six distances were measured among the five stations.

The a-priori standard deviations are:

- Angles: $\sigma_{\alpha,0} = 1.5''$
- Distances: $\sigma_{d,0} = 0.02m$

The observation covariance matrix is assumed diagonal:

$$C_{ll,0} = diag(\sigma_{\alpha,0}^2, \dots, \sigma_{\alpha,0}^2, \sigma_{d,0}^2, \dots, \sigma_{d,0}^2)$$

and P = C_{ll,0}^{-1}.

Using the same observation model and Gauss–Newton adjustment as Lab 1, the baseline solution yields:

- Adjusted coordinates (\hat{x} = [x_{P1}, y_{P1}, x_{P2}, y_{P2}]^\mathsf{T}) identical to Lab 1.
- Residuals (v̂) and residual cofactor matrix (Q_{vv}).
- A-posteriori variance factor:
  [
  \sigmâ_0^2 \approx 13.084.
  ]

Methodology

Q1 — Normality Test of GPS DD Noise

1. **Standardization**
   Each sample was standardized using sample mean and standard deviation:
   $$ti = sxi - \bar{x}$$

2. **Binning and Expected Frequencies**

   - The standardized range was partitioned into K = 22 class intervals in the t-domain, using a mixture of finite and open-ended bins:
     - First class: $(-\infty, t_1]$
     - Last class: $(t_{21}, +\infty)$
     - 20 interior classes defined by evenly spaced midpoints between -4 and $+4$.
   - For each class j, the observed frequency f_j was counted, and the theoretical expected frequency F_j was computed from the standard normal CDF:
     $$p_j = \Phi(b_j) - \Phi(a_j),\quad F_j = N p_j,$$
     where $(a_j, b_j]$ is the j-th interval.

3. **Chi-Square Goodness-of-Fit Test**
   The chi-square statistic was computed as:
   $$\chi^2 = \sum_{j=1}^{K} \frac{(f_j - F_j)^2}{F_j}.$$
   Because both the mean and variance are estimated, the degrees of freedom are:
   $$\nu = K - 3 = 19.$$
   For a confidence level of 95 %:
   $$\chi^2_{\text{crit}} = \chi^2_{0.95,\nu}.$$

4. **Graphical Diagnostics**

   - A histogram of the noise samples with overlaid normal PDF (using $(\bar{x}, s)$).
   - A Q–Q plot of the sorted samples against theoretical normal quantiles.

The decision is based primarily on the chi-square test, with the plots used to diagnose deviations (heavy tails, skewness, outliers).

Q2.1 — Global Variance-Factor Test and Local Data Snooping

Using the Lab 1 matrices from lab1_get_matrices():

- Number of observations: $n = 18$

- Number of unknowns: ($u = 4$)
- Degrees of freedom: ($r = n - u = 14$).

1. Global Variance-Factor Test
   With a-priori variance factor ($\sigma_{0,\text{apriori}}^2 = 1.0$), the test statistic is:
   $$Y = \frac{r\,\hat{\sigma}_0^2}{\sigma_{0,\text{apriori}}^2}.$$
   The 95 % acceptance interval is:
   $$\chi^2_{0.025,r} \le Y \le \chi^2_{0.975,r}.$$

2. Local Data Snooping (w-Tests)
   The cofactor of residuals is (Q_{vv}). For each observation (i):

   - Variance of residual:
     $$\sigma_{v_i}^2 = \hat{\sigma}_0^2\, q_{v_i v_i},$$
     where ($q_{v_i v_i}$) is the i-th diagonal element of (Q_{vv}).

   - Standardized w-statistic:
     $$w_i = \frac{v_i}{\sigma_{v_i}}.$$

   - Decision rule at 95 % confidence:
     $$|w_i| > K \quad\Rightarrow\quad \text{observation i flagged as blunder},$$
     with threshold ($K = \Phi^{-1}(0.95) \approx 1.645$).

Observations whose $|w|$ exceed K are considered blunder candidates.

Q2.2 — Correlation Coefficients Between Identification Statistics

To study the dependence between w-statistics, the correlation matrix of the residuals is formed from (Q_{vv}):

$$\rho_{ij} = \frac{q_{v_i v_j}}{\sqrt{q_{v_i v_i}\, q_{v_j v_j}}}, \quad i,j = 1,\dots,18.$$

In matrix form:

$$R = \left[\rho_{ij} \right] = D^{-1/2} Q_{vv} D^{-1/2},\quad D = \operatorname{diag}(q_{v_1v_1},\dots,q_{v_{18}v_{18}}).$$

This correlation matrix was visualized using a heatmap, with:

- Color scale fixed to [−1, 1].
- Observation indices (1–18) on the axes.
- Each cell annotated with the numeric value of (ρ_{ij}) to interpret the strength of correlation.

The maximum absolute off-diagonal correlation ($\max_{i\neq j} |\rho_{ij}|$) was reported to assess how strongly the tests are coupled.

Q3 — Interpretation When More Than Two Blunders Are Detected

If more than two observations were flagged as blunders, the analysis from Q2.1 and Q2.2 would be revisited to:

- Check whether large w-values come from one dominant blunder propagating through strongly correlated observations.
- Investigate whether systematic effects or model deficiencies (e.g., wrong standard deviations, neglected correlations) cause multiple tests to fail simultaneously.

In this dataset, exactly two observations were clearly identified as blunders, so this question is discussed conceptually.

Q4 — Removal of Identified Blunders and Variance Component Estimation

The two observations identified as blunders in Q2.1 are removed from the measurement list. A reduced observation set (16 observations) is formed by deleting those rows from:

- Observation vector l,
- Design matrix A,
- Weight matrix P.

A new adjustment is then performed with the same functional model to obtain:

- Updated residuals ($\hat{v}^{(\text{clean})}$),
- Updated ($Q_{vv}^{(\text{clean})}$),
- Updated variance factor ($\hat{\sigma}_{0,\text{clean}}^2$).

To estimate separate variances for angles and distances, a simplified variance component estimation approach is used:

- Partition the residuals into angular and distance groups.
- Use the diagonal elements of ($Q_{vv}^{(\text{clean})}$) and corresponding residuals to derive empirical variance estimates:
  [
  \hat{\sigma}*{\alpha}^2 \approx \text{scale factor} \times \text{average residual variance of angles},*
  ]
  [

$$\hat{\sigma}_{d}^2 \approx \text{scale factor} \times \text{average residual variance of distances}.$$

- Compare $\hat{\sigma}_{\alpha}$ and $\hat{\sigma}_{d}$ against the manufacturer specifications $1.5''$ and $0.02\ \text{m}$.

Q5–Q7 — Artificial Errors of 5σ and 3σ

A more general version of the Lab 1 helper was implemented, allowing controlled perturbation of selected distance measurements before adjustment:

$$d_k^{(\text{modified})} = d_k^{(\text{original})} + \Delta d_k,$$

where Δd_k is set to either $5\sigma_d$ or $3\sigma_d$ for one or two chosen distances.

Cases:

- Case A (Q5): Add $+5\sigma_d = +0.10\ \text{m}$ to distance P2–C.
- Case B (Q6): Add $+5\sigma_d$ to P2–C and P1–A simultaneously.
- Case C (Q7): Repeat Case A and Case B using $+3\sigma_d = +0.06\ \text{m}$ instead.

For each case:

1. Re-run the non-linear adjustment with the modified observations.
2. Compute the new a-posteriori variance factor $\hat{\sigma}_0^2$.
3. Perform global variance-factor test and local w-tests.
4. Recompute the correlation matrix of the residuals and plot the heatmap (the overall pattern is similar; emphasis is on how strengthened distance residuals show up in particular rows/columns).

Results & Discussion by Question

Q1 — Normality of GPS DD Noise

Histogram & Q–Q plot:

- The histogram shows a distribution that is more heavy-tailed than a Gaussian with the same mean and variance; several extreme values occur more often than expected.
- The Q–Q plot shows systematic departures from the straight line, with curvature at both ends, confirming significant departures from normality (heavy tails and possible skew).

Conclusion for Q1:
The double-differenced GPS noise in Lab4Q1data does not follow a perfect normal distribution under the i.i.d. normal assumption. This implies that using a simple Gaussian error model for these DD ranges may underestimate extreme noise events, and more robust modelling may be required.

Q2.1 — Global Variance-Factor Test & Data Snooping

From the Lab 1 network adjustment, the base solution gives:

- $n = 18$, $u = 4$, $r = 14$.

- A-posteriori variance factor:
$$\hat{\sigma}_0^2 \approx 13.084.$$
- Global test statistic:
$$Y = r\,\hat{\sigma}_0^2 \approx 14 \times 13.084 \approx 183.2.$$
- 95 % chi-square bounds for ($r = 14$):
- Lower: ($\chi^2_{0.025,14} \approx 5.63$)
- Upper: ($\chi^2_{0.975,14} \approx 26.12$).

Since:
$$Y \gg 26.12,$$
we strongly reject the hypothesis ($H_0 : \sigma_0^2 = 1.0$). This confirms the conclusion from Lab 1 that the a-priori stochastic model is too optimistic; the actual noise level is much larger than prescribed.

Local w-tests:

Using ($\hat{\sigma}_0^2 \approx 13.084$) and (Q_{vv}), the standardized residuals (w_i) were computed for all 18 observations. With threshold ($K \approx 1.645$) at 95 % confidence, the following observations were flagged:

- Observation 9: $|w_9| \approx 2.53 > 1.645$
- Observation 14: $|w_{14}| \approx 2.52 > 1.645$

All other $|w_i|$ are below the threshold.

Mapping back to the measurement list:

- Observation 9 → angle at P2 (A–P2–P1).
- Observation 14 → distance P1–B (second distance entry).

These two measurements are identified as likely blunders under the data snooping criterion.

Q2.2 — Correlation of Identification Statistics

From (Q_{vv}), the residual correlation matrix R was computed. The maximum off-diagonal absolute correlation was:

$$\max_{i\neq j} |\rho_{ij}| \approx 0.64.$$

The heatmap shows:

- Blocks of moderate correlation among angle residuals (due to shared stations and geometric coupling).

- Noticeable correlation between some distance residuals, particularly those involving P1 and P2, which share unknown coordinates (hence their residuals are not independent).

Implications:

- The w-tests assume, in theory, that the test statistics are independent standard normal variables under the correct model.
- In practice, moderate correlations ($|\rho| \sim 0.5$–0.6) mean that:
  - A large residual in one observation can propagate into neighboring observations via the adjustment, and
  - Multiple w-tests may become correlated, affecting the interpretation of multiple testing.

In this dataset, despite correlations, the two main blunders (obs 9 and 14) are clearly dominant and still detectable.

Q3 — Interpretation of Multiple Blunders

In the present network, exactly two observations (9 and 14) are clearly flagged at 95 % confidence. If more than two had been detected, likely explanations would include:

- **Single dominant blunder + correlation:**
  A very large error in one measurement can induce significant residuals in other, strongly correlated observations. The heatmap shows that several observations are moderately correlated, especially within angle groups and distance groups; thus multiple observations could cross the threshold even when only one is physically wrong.

- **Model misspecification:**
  If the stochastic model (variances, neglect of covariance, or functional model) is incorrect, residuals may be inflated or biased in a structured way, causing several w-tests to appear "bad" even without true blunders.

Because only two observations are clearly inconsistent here, the situation is consistent with two actual measurement blunders rather than a more fundamental model problem.

Q4 — Removing Blunders and Estimating Measurement Variances

After removing observations 9 and 14 from the data set and recomputing the residuals and variance factor (using the same functional model), the cleaned network yields:

- New degrees of freedom: $r_{\text{clean}} = 16 - 4 = 12$.
- New a-posteriori variance factor:
  $$\hat{\sigma}_{0,\text{clean}}^2 \approx 2.93.$$

This is much smaller than the original 13.084, showing that the two blunders were responsible for a large part of the global misfit. However, $\hat{\sigma}_{0,\text{clean}}^2$ is still greater than 1, so even after removing obvious blunders, the network noise is still higher than the optimistic a-priori model.

Using a simplified variance component interpretation:

- Effective standard deviation for angles:
$$\hat{\sigma}_\alpha \approx \sqrt{\hat{\sigma}_{0,\text{clean}}^2}\,\sigma_{\alpha,0} \approx \sqrt{2.93}\times 1.5'' \approx 2.6''.$$
- Effective standard deviation for distances:
$$\hat{\sigma}_d \approx \sqrt{\hat{\sigma}_{0,\text{clean}}^2}\,\sigma_{d,0} \approx \sqrt{2.93}\times 0.02\,\text{m} \approx 0.034\ \text{m}.$$

Comparison with manufacturer’s specification:

- Angles: Manufacturer: 1.5″ vs estimated ~2.6″ → instrument/field conditions produce larger angular noise.
- Distances: Manufacturer: 0.02 m vs estimated ~0.034 m → distance noise is also higher than specified.

Conclusion for Q4:
Removing the two most obvious blunders significantly improves the global variance factor, but the realistic noise level in this survey is still higher than the nominal specs. For practical work, it would be sensible to adopt the empirically estimated variances (or something between the a-priori and estimated values) for subsequent adjustments and reliability analyses.

Q5 — 5σ Error Added to Distance P2–C

In this scenario, a +5σ error (0.10 m) is added to the distance P2–C before adjustment:

$$d_{P2C}^{(\text{new})} = d_{P2C}^{(\text{orig})} + 0.10\ \text{m}.$$

The adjustment with this perturbed distance yields:

- $\hat{\sigma}_0^2$ increases from ≈13.084 to ≈15.76.
- The global chi-square statistic $Y = r\,\hat{\sigma}_0^2$ becomes even larger (farther outside the 95 % bounds), confirming the presence of additional gross error.

Local w-tests:

- Observations 9 and 14 remain flagged.
- The distance P2–C (obs 16) now also shows $|w_{16}| \approx 1.8 > 1.645$, and is therefore flagged as a blunder as well.

Interpretation:

- A 5σ error in P2–C is large enough that it cannot be absorbed by the network; it appears clearly in the residual of that distance and slightly perturbs related angles and distances.
- The data snooping procedure successfully detects and localizes this gross error (on top of the original two blunders).
- The correlation matrix shows stronger influence from the P2–C row/column, illustrating how a large distance error propagates into correlated observations but still manifests most strongly in its own residual.

Q6 — 5σ Errors Added to P2–C and P1–A

In this case, both distance measurements P2–C and P1–A receive +5σ perturbations:

$$
d_{P2C}^{(\text{new})} = d_{P2C}^{(\text{orig})} + 0.10\ \text{m},
$$

$$
d_{P1A}^{(\text{new})} = d_{P1A}^{(\text{orig})} + 0.10\ \text{m}.
$$

The resulting adjustment shows:

- A further increase in $\hat{\sigma}_0^2$ (network misfit grows even more).
- w-tests still clearly flag obs 9 and 14 (original angular and distance blunders) and obs 16 (P2–C).
- The residual for P1–A also increases; however, due to correlation and model geometry, its |w| is closer to the decision threshold and may be less dominant than the P2–C blunder.

Interpretation:

- When two correlated distances each contain a 5σ error, the network tries to reconcile both, spreading the misfit across several observations.
- Strong correlations between distances involving the same unknown points mean that a blunder in one distance can partially mask or share its impact with another, making it slightly harder to isolate each one purely from w-tests.
- Nevertheless, the overall global test strongly rejects the model, and at least one of the perturbed distances is clearly identified.

Q7 — Effect of 3σ Errors Instead of 5σ

Finally, the experiments in Q5 and Q6 were repeated using 3σ perturbations (0.06 m) instead of 5σ:

- Case A (3σ on P2–C only):

  - $\hat{\sigma}_0^2$ increases slightly relative to the base case but not as dramatically as in the 5σ case.
  - w-tests still primarily flag obs 9 and 14 as the dominant blunders.
  - The residual of P2–C is elevated but its |w| may be closer to the threshold, sometimes not exceeding 1.645, depending on rounding. The original blunders remain the most visible.

- Case B (3σ on P2–C and P1–A):

- The global variance factor is again slightly higher than in the base case.
- However, because both distance errors are “only” 3σ and the network is already stressed by existing blunders, the additional misfit is more easily absorbed by correlations.
- In the results, obs 9 and 14 continue to be the main outliers, while P1–A and P2–C may no longer be individually above the 95 % w-test threshold.

Summary of 3σ vs 5σ:

- 5σ errors are clearly detectable by data snooping and significantly inflate the variance factor; they stand out strongly in the w-statistics and are hard to hide in a correlated network.

- 3σ errors are more subtle: in the presence of other blunders and correlations, they may not always be singled out by w-tests, especially when combined with existing large errors. Their impact is visible in the global variance factor but less obvious in local tests.

This illustrates the limitations of data snooping at a fixed confidence level: moderate blunders (≈3σ) can be difficult to detect in the presence of multiple errors and correlated observations.

Conclusions

This lab demonstrated how statistical testing and variance component estimation complement classical least-squares adjustment:

1. **GPS DD Noise (Q1)**

    - The chi-square goodness-of-fit test and Q–Q plot showed that the DD noise samples deviate significantly from a normal distribution.
    - The noise is heavier-tailed than a Gaussian, indicating that occasional large deviations are more frequent than predicted by the simple normal model.

2. **Lab 1 Network — Global and Local Tests (Q2–Q3)**

    - The a-posteriori variance factor (\sigmâ_0^2 \approx 13.084) is far above 1.0, confirming that the a-priori noise model is too optimistic (as already seen in Lab 1).
    - Data snooping (w-tests) successfully identified two clear blunders (obs 9 and 14).
    - The correlation matrix revealed moderate correlations between residuals, meaning individual w-tests are not fully independent.

3. **Variance Component Estimation After Blunder Removal (Q4)**

    - Removing the two blunders reduced the variance factor to about 2.93, substantially improving model consistency but still above 1.
    - Estimated realistic standard deviations around 2.6″ for angles and 3.4 cm for distances are larger than manufacturer specs (1.5″ and 2 cm).
    - This suggests that either the instrument or the field environment (setup, atmosphere, centering, etc.) produces more noise than ideal, and the stochastic model should be updated accordingly.

4. **Artificial Errors and Detectability (Q5–Q7)**

- A 5σ error in a distance is clearly detectable by both the global chi-square test and local w-tests; adding 5σ to two distances further inflates the variance factor and flags multiple observations.
- 3σ errors are more subtle: in the presence of existing blunders and correlated observations, they may not always appear as standalone outliers in the w-tests, even though they slightly degrade the global fit.
- This highlights that data snooping at a fixed 95 % confidence level may miss moderate blunders, especially in correlated networks, and that complementary strategies (iterative reweighting, robust estimation, or alternative confidence levels) may be needed.

Overall, Lab 4 reinforced the idea that good geometry and a realistic stochastic model are both essential. Statistical tests help detect blunders and tune variances, but their power is limited by correlation and the chosen significance level.